

5. On detonation, the explosive RDX produces carbon monoxide, steam and nitrogen.



RDX  
 $M_r$  222

Calculate the mass of RDX that would give  $1.00\text{ m}^3$  of gaseous products measured at  $100^\circ\text{C}$  and at 1 atm. Give your answer to an **appropriate** number of significant figures. [2]

(1 mol of any gas occupies a volume of  $30.6\text{ dm}^3$  under these conditions)

Mass of RDX = ..... g

10

